Lesson 3: Unbiased Estimators, MVUE, Likelihood Theory

BSTA 551: Statistical Inference

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Lesson 3: Finding the Best Estimator

Review: Where We Left Off

Key concepts from Lesson 2:

- Point estimators vs. estimates
- Bias: $\mathrm{Bias}(\hat{\theta}) = E(\hat{\theta}) - \theta$
- Unbiasedness: $E(\hat{\theta}) = \theta$
- MSE = Variance + Bias^2

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Today's Goals:

- Understand what makes one unbiased estimator better than another
- Learn about Minimum Variance Unbiased Estimators (MVUE)
- Introduce likelihood functions (preview of Lesson 4)
- Define Fisher Information and its role in estimation
- Learn the Cramér-Rao Lower Bound for variance

Motivating Example: Estimating Drug Efficacy

A clinical trial tests the effectiveness of a new antibiotic. Two researchers propose different estimators for the true response rate p:

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Both estimators are functions of the same data. **Which should we use?**

```r
# Suppose 15 out of 20 patients respond
n <- 20; x <- 15
p_hat_A <- x / n
p_hat_B <- (x + 1) / (n + 2)
cat("Researcher A's estimate:", p_hat_A, "\n")
```

```
Researcher A's estimate: 0.75
```

```r
cat("Researcher B's estimate:", p_hat_B)
```

```
Researcher B's estimate: 0.7272727
```

Choosing Among Unbiased Estimators

The Problem of Multiple Unbiased Estimators

Recall from Lesson 2: There can be **many** unbiased estimators for the same parameter!

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Example: For a symmetric distribution with mean μ:

- Sample mean $\bar{X}$ is unbiased for μ
- Sample median $\tilde{X}$ is unbiased for μ
- Trimmed means are unbiased for μ

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- Sample mean $\bar{X}$ is unbiased for μ
- Sample median $\tilde{X}$ is unbiased for μ
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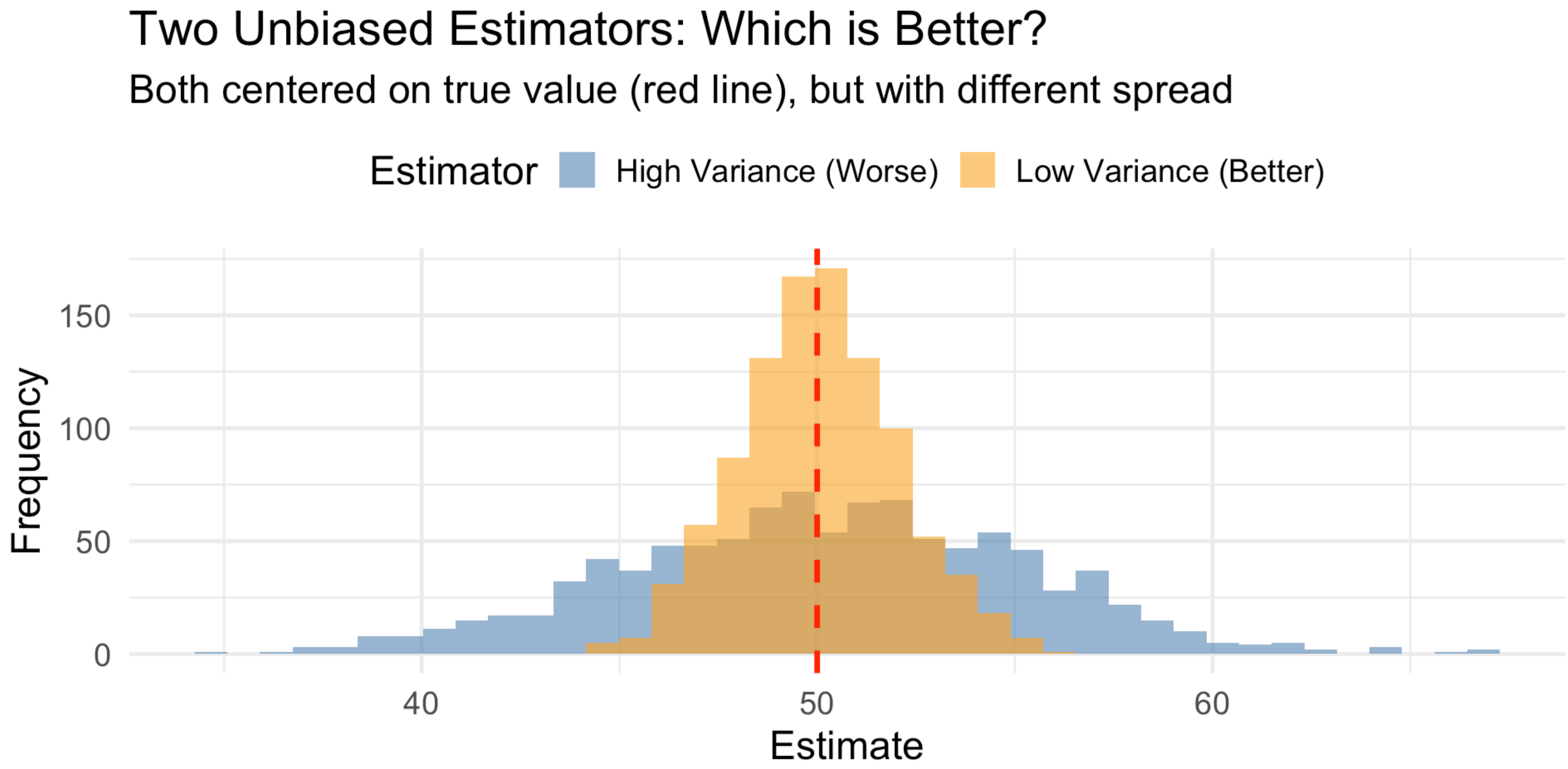
> **The Key Question**
>
> If two estimators are both unbiased, how do we choose between them?

Comparing Two Unbiased Estimators: Intuition



Answer: Among unbiased estimators, prefer the one with **smaller variance**.

The Principle: Minimum Variance Unbiased Estimator

Definition (Devore 7.1)

The **Minimum Variance Unbiased Estimator (MVUE)** of a parameter θ is the estimator $\hat{\theta}$ such that:

1. $\hat{\theta}$ is unbiased: $E(\hat{\theta}) = \theta$
2. $\hat{\theta}$ has smaller or equal variance than any other unbiased estimator

If $\hat{\theta}_1$ and $\hat{\theta}_2$ are both unbiased for θ, we prefer the one with smaller variance.

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Intuition: Among estimators that are "correct on average," we want the most precise one.

Example: Uniform Distribution Upper Bound

Setup: $X_1, \ldots, X_n \sim \mathrm{Uniform}[0, \theta]$

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2. $\hat{\theta}_2 = \frac{n+1}{n} \cdot \max(X_1, \ldots, X_n)$ (corrected maximum)

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Question: Which has smaller variance?

Calculating Variances

For $\hat{\theta}_1 = 2\bar{X}$:

$$\text{Var}(\hat{\theta}_1) = \text{Var}(2\bar{X}) = 4 \cdot \text{Var}(\bar{X}) = 4 \cdot \frac{\sigma^2}{n} = 4 \cdot \frac{\theta^2/12}{n} = \frac{\theta^2}{3n}$$

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Comparison: For $n > 1$:

$$\frac{\theta^2}{n(n+2)} < \frac{\theta^2}{3n}$$

The corrected maximum $\hat{\theta}_2$ is more efficient!

Simulation: Comparing the Two Estimators

```r
theta <- 10  # True upper bound
n <- 20
n_sims <- 5000

# Simulate both estimators
uniform_sim <- tibble(sim = 1:n_sims) |>
  mutate(
    data = map(sim, \(s) runif(n, 0, theta)),
    theta_hat_1 = map_dbl(data, \(d) 2 * mean(d)),
    theta_hat_2 = map_dbl(data, \(d) (n + 1) / n * max(d))
  )

# Compare variances
uniform_sim |>
  summarize(
    `Var(2*X̄)` = var(theta_hat_1),
    `Var(corrected max)` = var(theta_hat_2),
    `Theoretical Var(2*X̄)` = theta^2 / (3 * n),
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  pivot_longer(everything()) |>
  mutate(value = round(value, 4))
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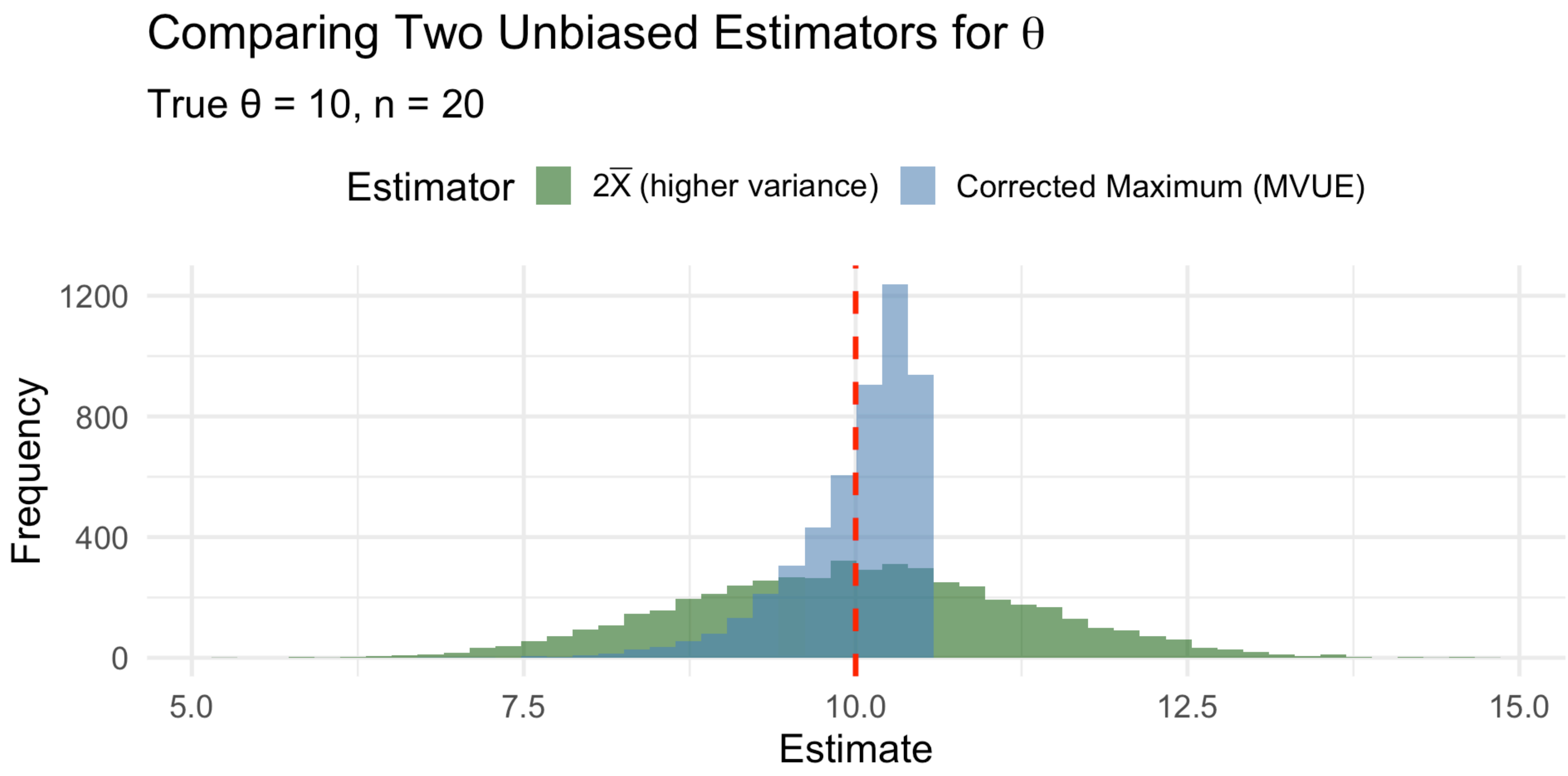
Visualizing the Comparison



The corrected maximum is **more concentrated** around the true value!

Introduction to Likelihood

The Likelihood Function: A Brief Introduction

Definition (Preview - Full treatment in Lesson 4)

The **likelihood function** $L(\theta)$ is the joint probability/density of the observed data, viewed as a *function of the parameter* θ:

$$L(\theta) = \prod_{i=1}^{n} f(x_i; \theta)$$

where f is the pmf (discrete) or pdf (continuous).

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Key Idea: The likelihood tells us how "likely" different parameter values are, given our observed data.

The Log-Likelihood Function

Taking the logarithm converts products to sums:

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- Numerically more stable
- Same maximum as likelihood (since log is monotonic)

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The **score function** is the first derivative: $\ell'(\theta) = \frac{d}{d\theta}\ell(\theta)$

Example: Bernoulli Likelihood

For $X_1, \ldots, X_n \sim \text{Bernoulli}(p)$ with $x = \sum x_i$ successes:

Likelihood:

$$L(p) = p^x (1-p)^{n-x}$$

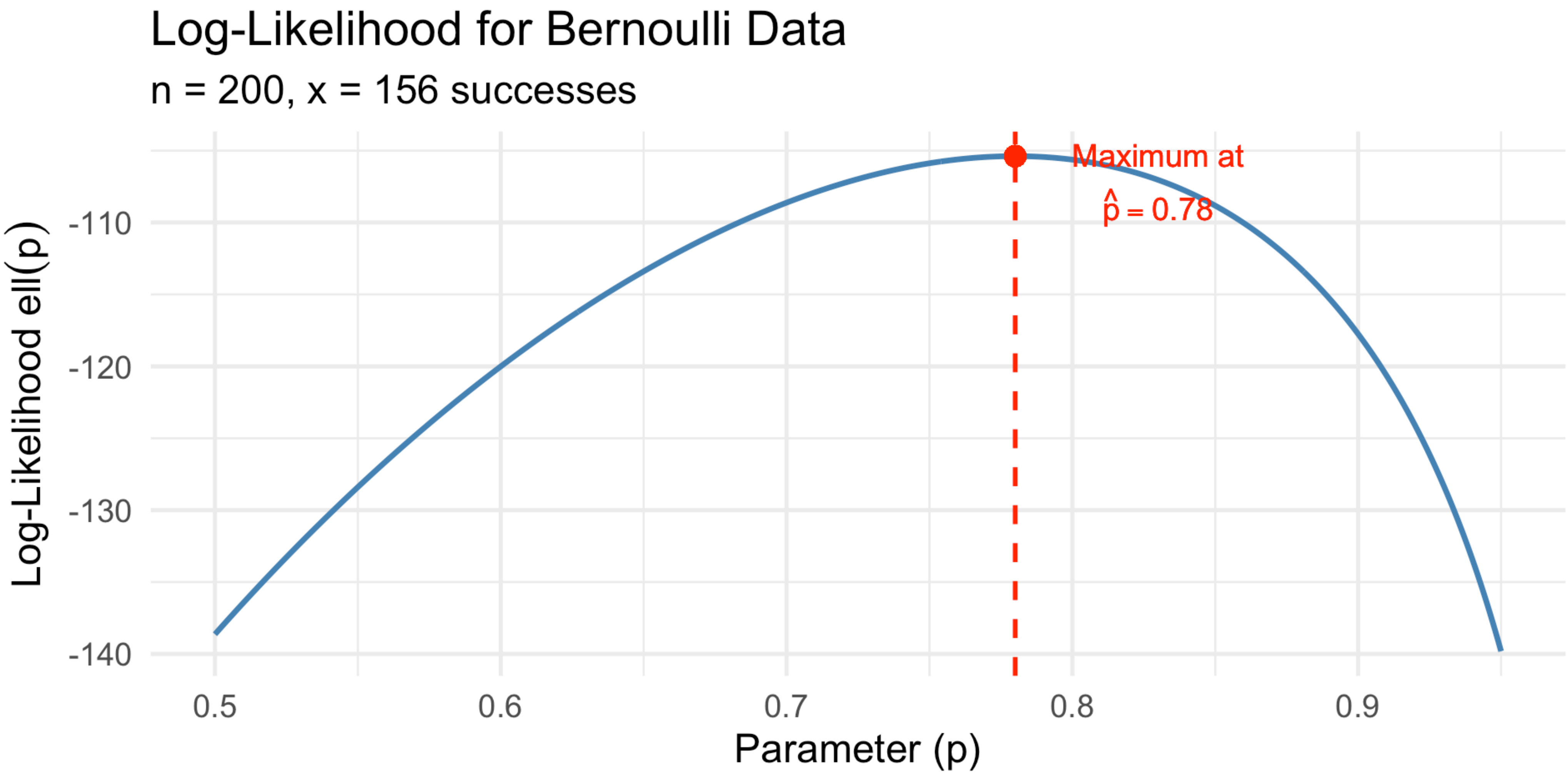
Log-likelihood:

$$\ell(p) = x \ln(p) + (n - x) \ln(1 - p)$$

Score function:

$$\ell'(p) = \frac{x}{p} - \frac{n-x}{1-p}$$

Visualizing the Log-Likelihood

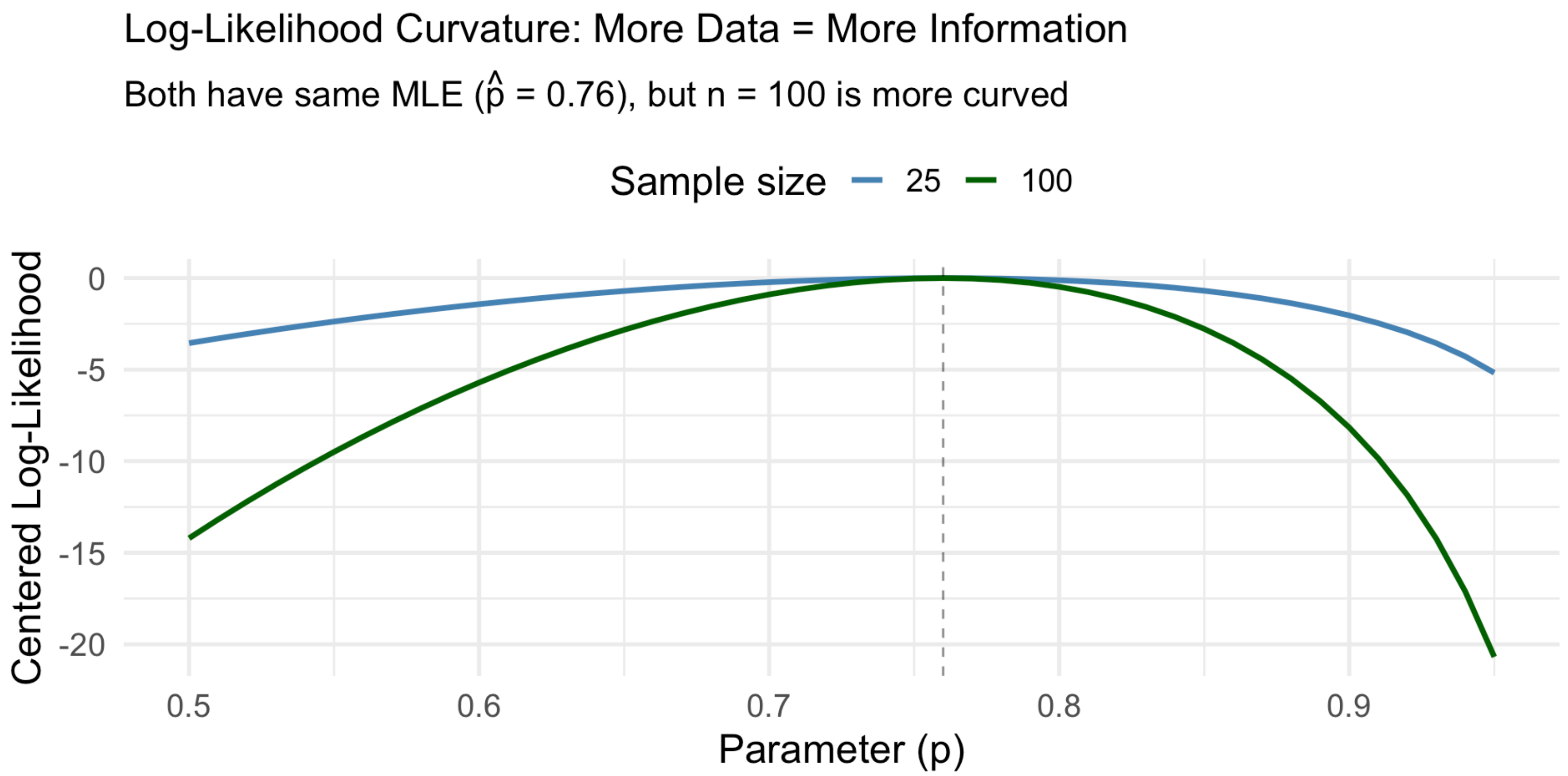


The **curvature** of this function contains information about how precisely we can estimate p.

Fisher Information

Curvature and Information

Key Observation: The more curved the log-likelihood is at its peak, the more precisely we can estimate the parameter.



More curvature = more information = more precise estimates

Fisher Information: Definition (Devore 7.4)

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Let $f(x;\theta)$ be a pmf or pdf. The **Fisher Information** in a single observation X is:

$$I(\theta) = E\left[-\frac{\partial^2}{\partial\theta^2}\ln f(X;\theta)\right] = E[-\ell''(\theta)]$$

Equivalently:

$$I(\theta) = E\left[\left(\frac{\partial}{\partial\theta}\ln f(X;\theta)\right)^2\right] = E[(\ell'(\theta))^2] = \mathrm{Var}(\ell'(\theta))$$

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Intuition: Fisher information measures the expected curvature of the log-likelihood.

Fisher Information in a Sample

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Key Properties

- $I(\theta) > 0$ (always positive)
- Larger sample size → more information
- Information depends on the distribution, not the data

Example: Fisher Information for Bernoulli

For $X \sim \text{Bernoulli}(p)$: $f(x;p) = p^x(1-p)^{1-x}$

Step 1: Log-likelihood for single observation:

$$\ln f(x;p) = x\ln(p) + (1-x)\ln(1-p)$$

Step 2: Second derivative:

$$\frac{\partial^2}{\partial p^2}\ln f(x;p) = -\frac{x}{p^2} - \frac{1-x}{(1-p)^2}$$

Bernoulli Fisher Information (continued)

Step 3: Take expectation (multiply by -1):

$$I(p) = E\left[\frac{X}{p^2} + \frac{1-X}{(1-p)^2}\right]$$

$$= \frac{E(X)}{p^2} + \frac{1-E(X)}{(1-p)^2} = \frac{p}{p^2} + \frac{1-p}{(1-p)^2}$$

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For a sample of size n:

$$I_n(p) = \frac{n}{p(1-p)}$$

Example: Fisher Information for Normal (known σ)

For $X \sim N(\mu, \sigma^2)$ with σ known:

$$f(x;\mu) = \frac{1}{\sqrt{2\pi}\sigma} e^{-(x-\mu)^2/(2\sigma^2)}$$

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$$I(\mu) = E\left[\frac{1}{\sigma^2}\right] = \frac{1}{\sigma^2}$$

For a sample: $I_n(\mu) = \frac{n}{\sigma^2}$

Example: Fisher Information for Exponential

For $X \sim \text{Exponential}(\lambda)$: $f(x; \lambda) = \lambda e^{-\lambda x}$

Log-likelihood: $\ln f(x; \lambda) = \ln(\lambda) - \lambda x$

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Derivatives:

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Your Turn: Calculate Fisher Information

Exercise: For the Poisson distribution with parameter μ, where $f(x;\mu) = \frac{e^{-\mu}\mu^x}{x!}$:

1. Find $\ln f(x;\mu)$
2. Calculate $\frac{\partial^2}{\partial\mu^2} \ln f(x;\mu)$
3. Find $I(\mu)$

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Solution:

1. $\ln f = -\mu + x\ln(\mu) - \ln(x!)$
2. $\frac{\partial}{\partial\mu}\ln f = -1 + \frac{x}{\mu}$, $\frac{\partial^2}{\partial\mu^2}\ln f = -\frac{x}{\mu^2}$
3. $I(\mu) = E\left[\frac{X}{\mu^2}\right] = \frac{E(X)}{\mu^2} = \frac{\mu}{\mu^2} = \frac{1}{\mu}$

The Cramér-Rao Lower Bound

The Fundamental Inequality

Cramér-Rao Inequality (Devore 7.4)

Let $X_1, \ldots, X_n$ be a random sample from $f(x;\theta)$ whose support does not depend on θ. If $T = t(X_1, \ldots, X_n)$ is an **unbiased** estimator of θ, then:

$$\mathrm{Var}(T) \geq \frac{1}{I_n(\theta)} = \frac{1}{n \cdot I(\theta)}$$

The right-hand side is the **Cramér-Rao Lower Bound (CRLB)**.

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The right-hand side is the **Cramér-Rao Lower Bound (CRLB)**.

In words: No unbiased estimator can have variance smaller than $1/I_n(\theta)$.

Why the CRLB Matters

The Cramér-Rao bound tells us:

1. **The best possible precision** for any unbiased estimator
2. **Whether an estimator is efficient** (achieves the bound)
3. **How much we lose** by using a suboptimal estimator

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Efficiency Definition

An unbiased estimator T is **efficient** if:

$$\mathrm{Var}(T) = \frac{1}{I_n(\theta)}$$

An efficient estimator is the MVUE!

Example: Is $\hat{p}$ Efficient for Bernoulli?

For $X_1, \ldots, X_n \sim \text{Bernoulli}(p)$:

Estimator: $\hat{p} = \bar{X} = \frac{\sum X_i}{n}$

Variance: $\text{Var}(\hat{p}) = \frac{p(1-p)}{n}$

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$\hat{p}$ is efficient and is the MVUE of p!

Example: Is $\bar{X}$ Efficient for Normal Mean?

For $X_1, \ldots, X_n \sim N(\mu, \sigma^2)$ with known σ:

Estimator: $\bar{X}$

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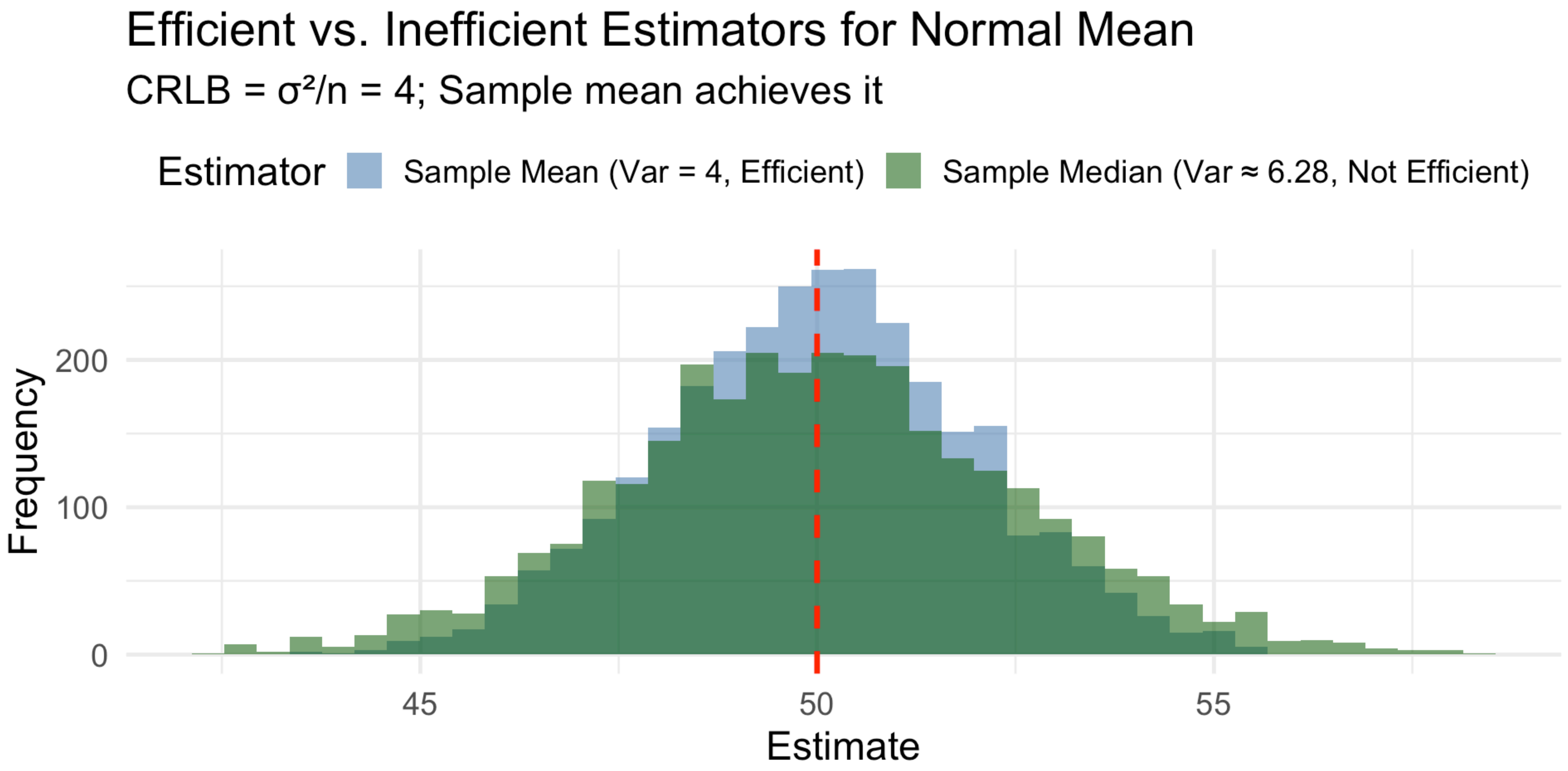
Visualizing Efficiency



Efficiency of the Sample Median

For normal data, the sample median is unbiased for μ but has:

$$\mathrm{Var}(\tilde{X}) \approx \frac{\pi}{2} \cdot \frac{\sigma^2}{n} \approx 1.57 \cdot \frac{\sigma^2}{n}$$

Efficiency of the Sample Median

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$$\mathrm{Var}(\tilde{X}) \approx \frac{\pi}{2} \cdot \frac{\sigma^2}{n} \approx 1.57 \cdot \frac{\sigma^2}{n}$$

Efficiency of median:

$$\text{Efficiency} = \frac{\text{CRLB}}{\mathrm{Var}(\tilde{X})} = \frac{\sigma^2/n}{1.57 \cdot \sigma^2/n} \approx 0.64$$

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The median is only about **64% as efficient** as the mean for normal data.

We need about **1.57 times as many observations** with the median to achieve the same precision as the mean!

Medical Application: Estimating Treatment Effect

A clinical trial measures systolic BP reduction (mmHg). Assume $N(\mu, \sigma^2 = 100)$.

```r
n <- 30
sigma_sq <- 100

# CRLB for μ
crlb <- sigma_sq / n

# Variance of sample mean
var_xbar <- sigma_sq / n

# Efficiency check
cat("CRLB for μ:", round(crlb, 3), "\n")
```

```
CRLB for μ: 3.333
```

```r
cat("Var(X̄):", round(var_xbar, 3), "\n")
```

```
Var(X̄): 3.333
```

```r
cat("Is X̄ efficient?", var_xbar == crlb, "\n\n")
```

```
Is X̄ efficient? TRUE
```

```r
# Sample data
bp_data <- c(12.3, 8.7, 15.2, 10.1, 7.4, 14.8, 11.2, 9.5, 13.1, 16.2,
             8.1, 12.5, 10.8, 14.3, 11.7, 9.2, 15.0, 13.4, 7.9, 12.1,
             10.5, 8.3, 14.1, 11.5, 13.8, 9.8, 11.9, 12.7, 10.3, 14.6)
```

When is the CRLB Not Attained?

Not all unbiased estimators achieve the CRLB!

Example: Uniform[0, θ] distribution

- Fisher Information: $I(\theta) = 1/\theta^2$
- CRLB: θ^2/n

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Two unbiased estimators:

- $\hat{\theta}_1 = 2\bar{X}$ has variance $\theta^2/(3n)$
- $\hat{\theta}_2 = \frac{n+1}{n} \max(X_i)$ has variance $\theta^2/[n(n+2)]$

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Neither achieves CRLB = θ^2/n! Why?

The support of Uniform[0, θ] depends on θ, violating a regularity condition.

When Does the CRLB Not Apply?

The Cramér-Rao inequality requires:

1. The **support** of $f(x;\theta)$ doesn't depend on θ
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Examples where CRLB doesn't apply:

- Uniform[0, θ] - support depends on θ
- Uniform[$\theta, \theta + 1$] - support depends on θ
- Some discrete distributions with restricted parameter spaces

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⚠ **Warning**

Always check regularity conditions before applying the CRLB!

MVUE Summary

Finding the MVUE

1. Calculate $I(\theta)$ for the distribution
2. Compute CRLB = $1/[n \cdot I(\theta)]$
3. Find an unbiased estimator
4. Check if its variance equals the CRLB
5. If yes → it's the MVUE!

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Key Results:

Distribution	Parameter	MVUE	CRLB
Bernoulli(p)	p	$\bar{X}$	$p(1-p)/n$
Normal(μ, σ^2)	μ	$\bar{X}$	σ^2/n
Exponential(λ)	λ	See note	λ^2/n
Poisson(μ)	μ	$\bar{X}$	μ/n

Summary and Looking Ahead

Lesson 3 Summary

Key Concepts:

1. **MVUE:** The unbiased estimator with minimum variance
   - Among unbiased estimators, prefer smaller variance
2. **Likelihood & Log-Likelihood:**
   - $L(\theta) = \prod f(x_i; \theta), \ell(\theta) = \sum \ln f(x_i; \theta)$
   - Curvature relates to estimation precision
3. **Fisher Information:** $I(\theta) = E[-\ell''(\theta)]$
   - Measures curvature of log-likelihood
   - Additive across observations: $I_n(\theta) = n \cdot I(\theta)$
4. **Cramér-Rao Lower Bound:**
   - $\mathrm{Var}(T) \geq 1/I_n(\theta)$ for unbiased T
   - Efficient estimators achieve the bound → MVUE

Key Formulas to Remember

Distribution	Parameter	$I(\theta)$	CRLB
Bernoulli	p	$\frac{1}{p(1-p)}$	$\frac{p(1-p)}{n}$
Normal (known σ)	μ	$\frac{1}{\sigma^2}$	$\frac{\sigma^2}{n}$
Exponential	λ	$\frac{1}{\lambda^2}$	$\frac{\lambda^2}{n}$
Poisson	μ	$\frac{1}{\mu}$	$\frac{\mu}{n}$

Lesson 3 Practice Problems

1. **(Devore 7.51)** For the geometric distribution $P(X = x) = (1 - p)^{x-1}p$, find Fisher information and the CRLB.
2. **(Devore 7.52)** For Poisson(μ), show that $\bar{X}$ is an efficient estimator.
3. **(Devore 7.53)** For Uniform[0, θ], explain why neither $2\bar{X}$ nor the corrected maximum achieves the CRLB.
4. **Medical Application:** In a trial, 72 of 90 patients respond. Calculate Fisher information, CRLB, and verify that $\hat{p} = 72/90$ is efficient.

Next Lesson Preview

Lesson 4: Method of Moments and Maximum Likelihood Estimation

- Method of Moments: Equate population and sample moments
- Maximum Likelihood: Maximize the likelihood function
- Properties: invariance, consistency, asymptotic normality
- MLEs are asymptotically efficient!

References

- Devore, Berk, and Carlton. *Modern Mathematical Statistics with Applications* (Springer). Chapters 7.1, 7.4
- Chihara and Hesterberg. *Mathematical Statistics with Resampling and R* (Wiley). Chapter 6.

Questions?

Thank you!